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<tr><th rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks Available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>7</td><td>Indicative content<br>In systemic circulation:<br>• {High pressure/ pressure increase} (to 120 mmHg) in {aorta/ arteries} is due to ventricular systole/ contraction of left ventricle.<br>• High pressure needed to pump blood {round body/ long distance}<br>• Pressure decrease (to 80 mmHg) in {arteries / aorta} is due to {diastole/ ventricular relaxation}.<br>• High pressure is maintained due to elastic recoil of artery walls / closure of {semilunar / aortic} valve.<br>• Pressure decrease in {arterioles/ capillaries} is due to {frictional resistance / blood flowing through a greater (total) {cross sectional/ surface area}.<br>• Pressure decrease in capillaries is also due to tissue fluid formation.<br>• low pressure/ flow rate in capillaries necessary for diffusion<br>• Blood flow in veins due to massaging effect of skeletal muscles.<br>In pulmonary circulation:<br>• Pressure increase in pulmonary arteries due to {ventricular systole/ contraction of thinner walled right ventricle}.<br>• Lower pressure (than systemic) due to {shorter distance / (only) to lungs}.<br>• Pressure decrease (to about 8 mmHg) in pulmonary arteries is due to diastole / ventricle relaxation.<br>• Pressure in capillaries is lower than systemic to prevent tissue fluid formation in the alveoli / reduce damage to alveoli / allow time for gas exchange.</td><td>6</td><td>3</td><td></td><td>9</td><td></td><td></td></tr>
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